Activity 1 — Data Science with Google Colab

Step 1 — Answer the class intake form first. Before any Colab work, answer the class intake & prior-knowledge form: **https://forms.gle/Z2AJpKTMx5CZcdFq7** (opens in a new tab). *Sagutan muna ito bago magsimula.* Everything below is **Step 2 onward**.

What you'll learn

You'll connect **your own Google Form survey** to Google Colab and turn the responses into charts — the same thing a data analyst does, just with smaller data. By the end you'll have bar charts, pie charts, a comparison between two questions, and a full survey dashboard built from data **you** collected.

Before you start

- A **Google account** (the same one you'll use in Colab).
- A short **Google Form** (4–6 questions, mostly multiple choice) with a **linked response Sheet**.
- **5–6 responses** in your form, so your charts aren't empty. *Pwede kayong mag-sagutan ng katabi nyo.*
- Your **Sheet ID** (you'll copy it from the Sheet's URL — see Step 4).

Make your form first. Go to **forms.google.com** → create a form → Responses tab → green Sheets icon → **Create spreadsheet**. Fill it out 5–6 times. Use the **same Google account** that owns the form when you log into Colab — *magkaiba ang account, hindi gagana.*

Run each cell in Colab by clicking it and pressing **Shift + Enter**. Go in order, top to bottom.

Step 1 — Install the tools

This installs the libraries that let Colab talk to Google Sheets. Run it once.

COPY THIS CODE

```
!pip install gspread google-auth
```

You should see: installation messages, ending with no errors (or "Requirement already satisfied").

Step 2 — Load the libraries

This brings in pandas (tables), matplotlib (charts), and the Google connection tools.

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```
import pandas as pd
import matplotlib.pyplot as plt
import gspread
from google.auth import default
from google.colab import auth

print("Setup complete!")
```

You should see: `Setup complete!`

Step 3 — Log in to Google

A popup will ask you to choose your Google account and allow access. Pick the account that **owns your form**, and click through every **Allow** / **Continue** screen.

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```
auth.authenticate_user()
creds, _ = default()
gc = gspread.authorize(creds)

print("Connected to Google!")
print("Ready to open your Google Sheet.")
```

Popup hiding? It sometimes appears behind the window or gets blocked. Check your address bar for a blocked-popup icon, or look behind the browser. *Sariling account nyo, hindi ang account ng trainer.*

You should see: `Connected to Google!`

Step 4 — Connect to YOUR sheet

Open your linked Google Sheet and look at the URL:

`https://docs.google.com/spreadsheets/d/`**1AbCdEf...1234567890**`/edit`

The **bold part between `/d/` and `/edit`** is your Sheet ID. Paste it between the quotes below.

> **The ID below is only an example placeholder** — it will fail until you replace it with **your own** Sheet ID.

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```
# ==== PASTE YOUR OWN SHEET ID BETWEEN THE QUOTES ====
SHEET_ID = "PASTE_YOUR_SHEET_ID_HERE"
# =====================================================

# Connect and pull the data once
try:
    sheet = gc.open_by_key(SHEET_ID).get_worksheet(0)
    data = sheet.get_all_values()

    if len(data) > 1:
        df = pd.DataFrame(data[1:], columns=data[0])
        print(f"SUCCESS! Connected to your survey.")
        print(f"Responses found: {len(df)}")
        print(f"Questions (columns): {len(df.columns)}")
        print("\nYour questions:")
        for i, col in enumerate(df.columns):
            short = col[:60] + "..." if len(col) > 60 else col
            print(f"  Column {i}: {short}")
    else:
        print("Connected, but no responses yet.")
        print("Go fill out your form a few times, then run this cell again.")

except Exception as e:
    print(f"Connection failed: {e}")
    print("\nCheck these:")
    print("1. Is your SHEET_ID pasted correctly (no extra spaces)?")
    print("2. Are you logged in with the account that owns the form?")
    print("3. Did you run Step 3 first?")
```

> **You should see:** `SUCCESS!`, the number of responses, and a numbered list of your questions. Remember those **Column numbers** — you'll use them to pick which question to chart.

Step 5 — See your full data table

Shows everything you collected as a clean table.

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```
if len(data) > 1:
    print(f"Rows (responses): {len(df)}")
    print(f"Columns (questions): {len(df.columns)}")
    print("\nYour complete data:")
    from IPython.display import display
    display(df)
else:
    print("No responses yet. Fill out your form first, then re-run Step 4.")
```

> **You should see:** your full survey as a table.

Step 6 — Bar chart of one question

Counts how many people picked each answer. Change `question_column` to chart a different question (use the Column numbers from Step 4).

COPY THIS CODE

```
question_column = 1   # change this number to chart a different question

if len(data) > 1 and question_column < len(df.columns):
    question_name = df.columns[question_column]
    answer_counts = df.iloc[:, question_column].value_counts()

    print(f"Bar chart for: {question_name}")

    plt.figure(figsize=(12, 6))
    bars = plt.bar(answer_counts.index.astype(str), answer_counts.values)

    for bar, count in zip(bars, answer_counts.values):
        plt.text(bar.get_x() + bar.get_width()/2, bar.get_height() + 0.05,
                 str(count), ha='center', va='bottom', fontweight='bold')

    plt.title(f'{question_name}', fontsize=14, fontweight='bold')
    plt.xlabel('Answers')
    plt.ylabel('Number of people')
    plt.xticks(rotation=45, ha='right')
    plt.grid(axis='y', alpha=0.3)
    plt.tight_layout()
    plt.show()

    print(f"Most popular answer: '{answer_counts.index[0]}' ({answer_counts.iloc[0]}
people)")
else:
    print("No data, or that question number does not exist. Check Step 4.")
```

You should see: a bar chart with counts on each bar. **Try it:** change the number to 2 and re-run — that one line is the whole trick.

Step 7 — Pie chart of one question

Same counts shown as percentages of the whole.

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```
question_column = 2   # change this number to chart a different question

if len(data) > 1 and question_column < len(df.columns):
    question_name = df.columns[question_column]
    answer_counts = df.iloc[:, question_column].value_counts()

    print(f"Pie chart for: {question_name}")

    plt.figure(figsize=(8, 6))
    plt.pie(answer_counts.values,
            labels=answer_counts.index.astype(str),
            autopct=lambda p: f'{p:.0f}%\n({int(round(p*answer_counts.sum()/100))})',
            startangle=90)
    plt.title(f'{question_name}', fontsize=14, fontweight='bold')
    plt.axis('equal')
    plt.tight_layout()
    plt.show()

    print(f"Most people answered: '{answer_counts.index[0]}'")
else:
    print("No data, or that question number does not exist. Check Step 4.")
```

You should see: a pie chart with percentages and counts.

Step 8 — Compare two questions (crosstab)

A crosstab shows how answers to one question relate to another, as a grouped bar chart.

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```
question_a = 1   # first question
question_b = 2   # second question

if len(data) > 1 and question_a < len(df.columns) and question_b < len(df.columns):
    name_a = df.columns[question_a]
    name_b = df.columns[question_b]

    # Build the crosstab: counts of each combination
    crosstab = pd.crosstab(df.iloc[:, question_a], df.iloc[:, question_b])

    print(f"How '{name_a}' relates to '{name_b}':\n")
    display(crosstab)

    crosstab.plot(kind='bar', figsize=(12, 6))
    plt.title(f'{name_a}  vs  {name_b}', fontsize=13, fontweight='bold')
    plt.xlabel(name_a)
    plt.ylabel('Number of people')
    plt.xticks(rotation=45, ha='right')
    plt.legend(title=name_b, bbox_to_anchor=(1.02, 1), loc='upper left')
    plt.grid(axis='y', alpha=0.3)
    plt.tight_layout()
    plt.show()

    print("Each group of bars is one answer to the first question.")
    print("The colors show how those people answered the second question.")
else:
    print("No data, or one of those question numbers does not exist. Check Step 4.")
```

You should see: a table and a grouped bar chart comparing two questions.

Step 9 — Full dashboard

Puts a pie chart for each question into one picture — a quick overview of your whole survey.

COPY THIS CODE

```python
if len(data) > 1:
    question_cols = list(range(1, len(df.columns)))  # skip timestamp at column 0
    n = len(question_cols)

    cols = 3
    rows = (n + cols - 1) // cols
    fig, axes = plt.subplots(rows, cols, figsize=(6*cols, 5*rows))
    fig.suptitle('Survey Results Dashboard', fontsize=16, fontweight='bold')

    axes_flat = axes.flatten() if n > 1 else [axes]

    for i, qcol in enumerate(question_cols):
        counts = df.iloc[:, qcol].value_counts()
        axes_flat[i].pie(counts.values,
                         labels=counts.index.astype(str),
                         autopct='%1.0f%%',
                         startangle=90)
        title = df.columns[qcol]
        title = title[:30] + "..." if len(title) > 30 else title
        axes_flat[i].set_title(title, fontsize=10, fontweight='bold')

    # turn off any leftover empty boxes
    for j in range(n, len(axes_flat)):
        axes_flat[j].axis('off')

    plt.tight_layout()
    plt.show()
    print(f"Dashboard created for {n} questions.")
else:
    print("No data to build a dashboard yet. Fill out your form first.")
```

You should see: a dashboard of pie charts, one per question.

Tips & common errors

- Run cells **top to bottom** with Shift + Enter. If Step 4 fails, you probably skipped Step 3.
- **Connection failed?** Check the Sheet ID for typos/extra spaces, and confirm you logged in with the account that **owns the form**.
- **Empty charts?** You have no responses yet — fill out your form, then re-run Step 4.

- To chart a different question, just change the **question number** and re-run that cell.

What's next

In **Activity 2** you'll collect data a different way — by **scraping it off the web** with code.
